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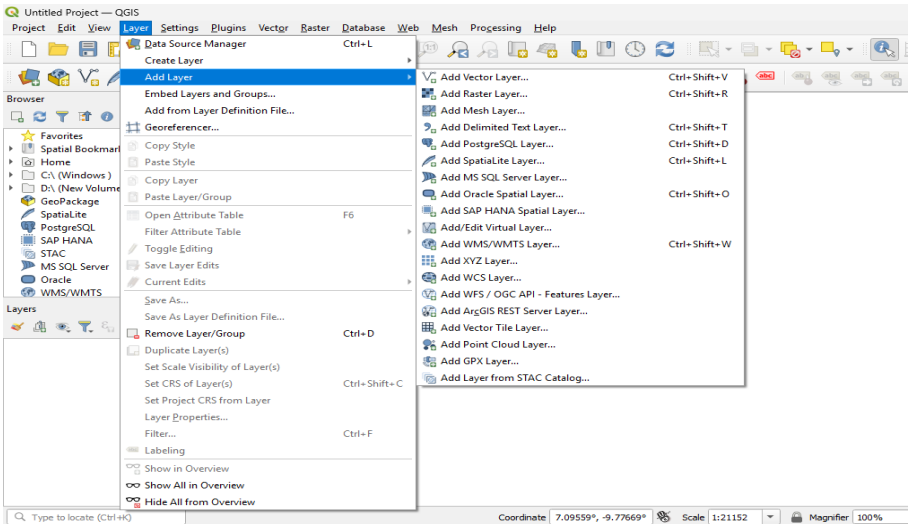
Q] Exploring and Managing Raster Data:

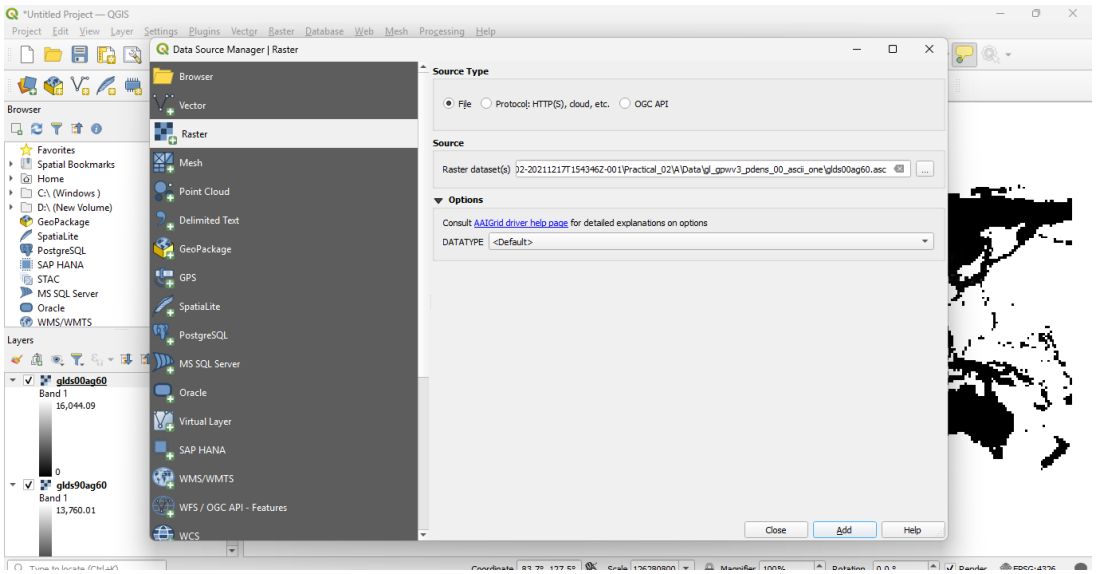
A. Adding raster layers:

Steps:

1. Add Raster Layer.



2. Browse and select the file glds90ag60.asc, then click the Add button.
   Similarly, add the file glds00ag60.asc.



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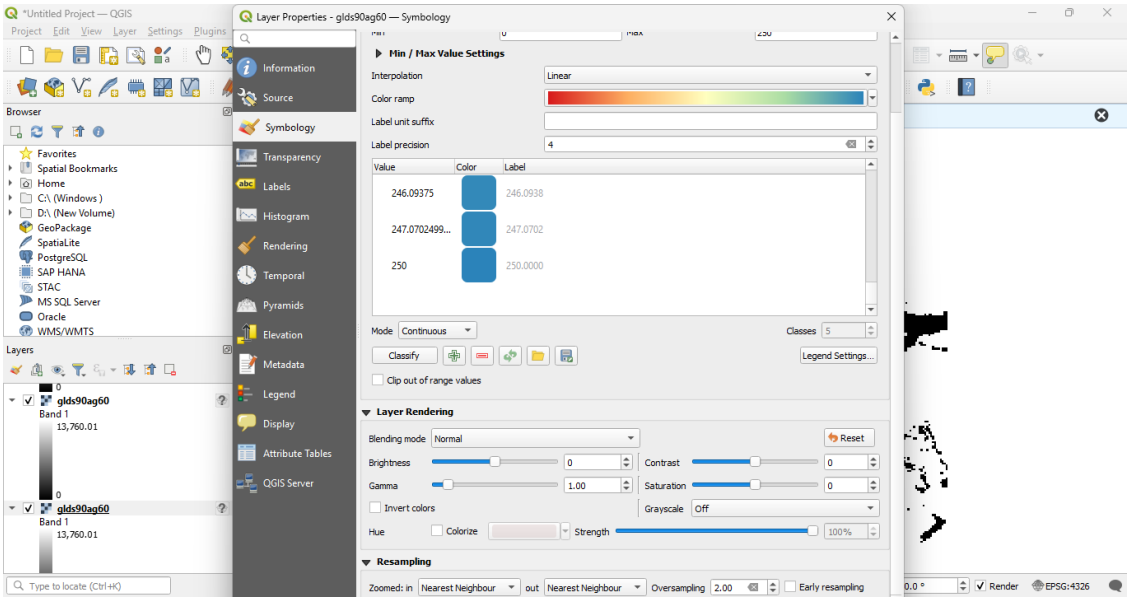
B. Raster Styling and Analysis:

1. Right-click on the glds90ag60 layer and select Properties.
2. Go to Symbology. In Render type, select Singleband pseudocolor.
   Set:
   - Min value: 0
   - Max value: 240
   - Under Interpolation, select Linear.
   - In the Color ramp menu, choose Spectral.
   - Click Apply, then click OK.



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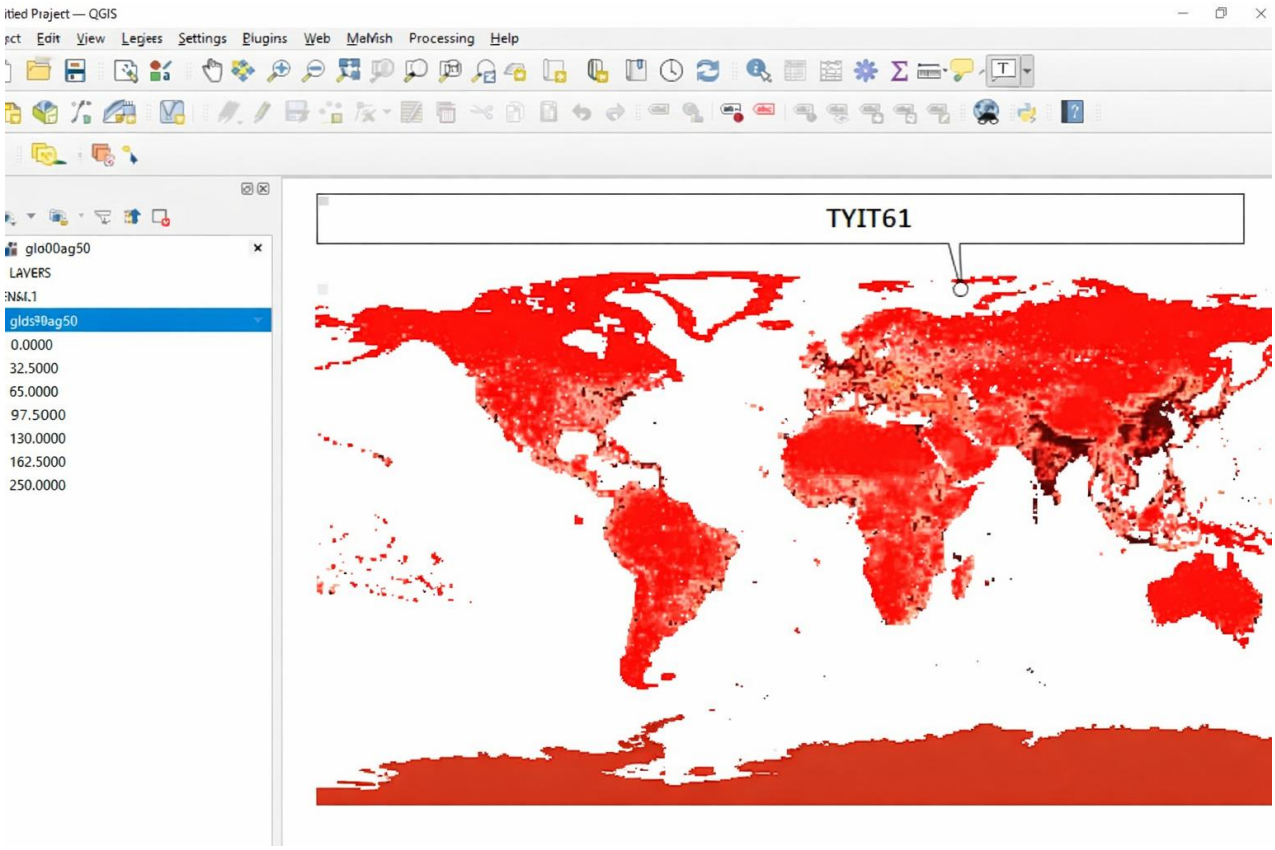
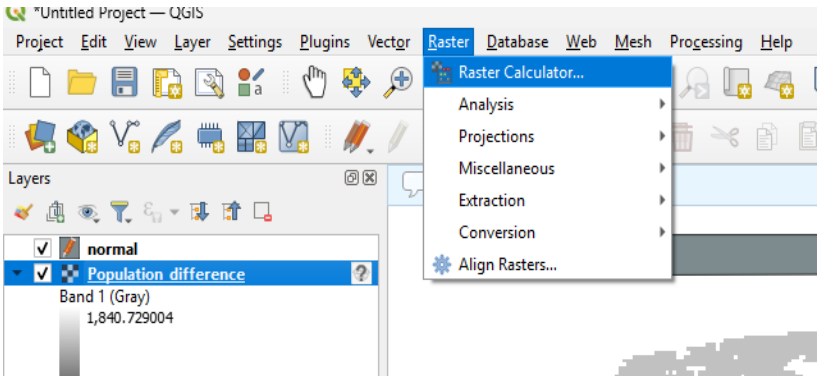


3. Go to Raster → Raster Calculator.



4. Click on the … (Browse) button. A menu will appear—select the path where you want to save the output. Enter a name for the output file and click Save.
   In the Raster Calculator window:
   - Double-click on glds00ag60@1
   - Click on the "-" (minus) operator

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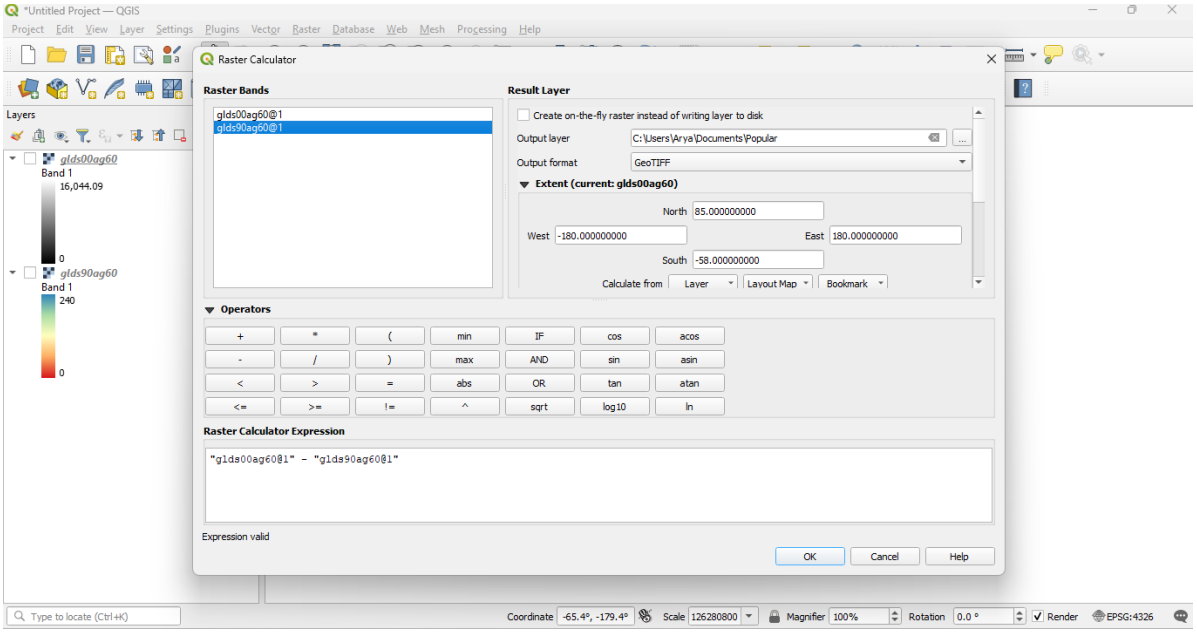
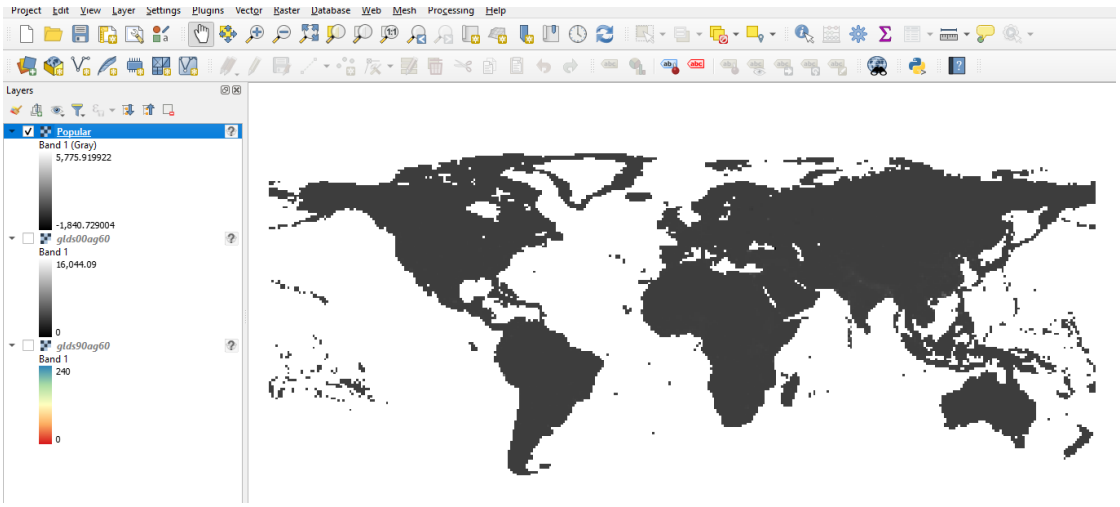
- Then double-click on glds90ag60@1



5. Click OK to generate a new raster layer.



6. Right-click on the newly created layer and select Properties -> Go to the Symbology tab. -> Set Render type to Single band pseudocoelom.
   Set :
   - Minimum value to –20000 and Maximum value to 6000.
   - Set Interpolation to Discrete.

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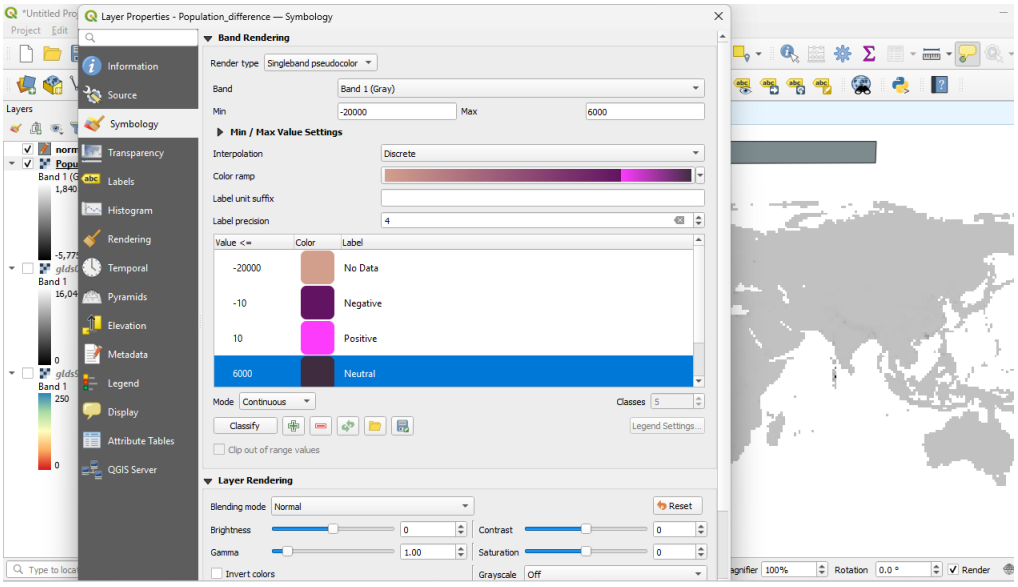
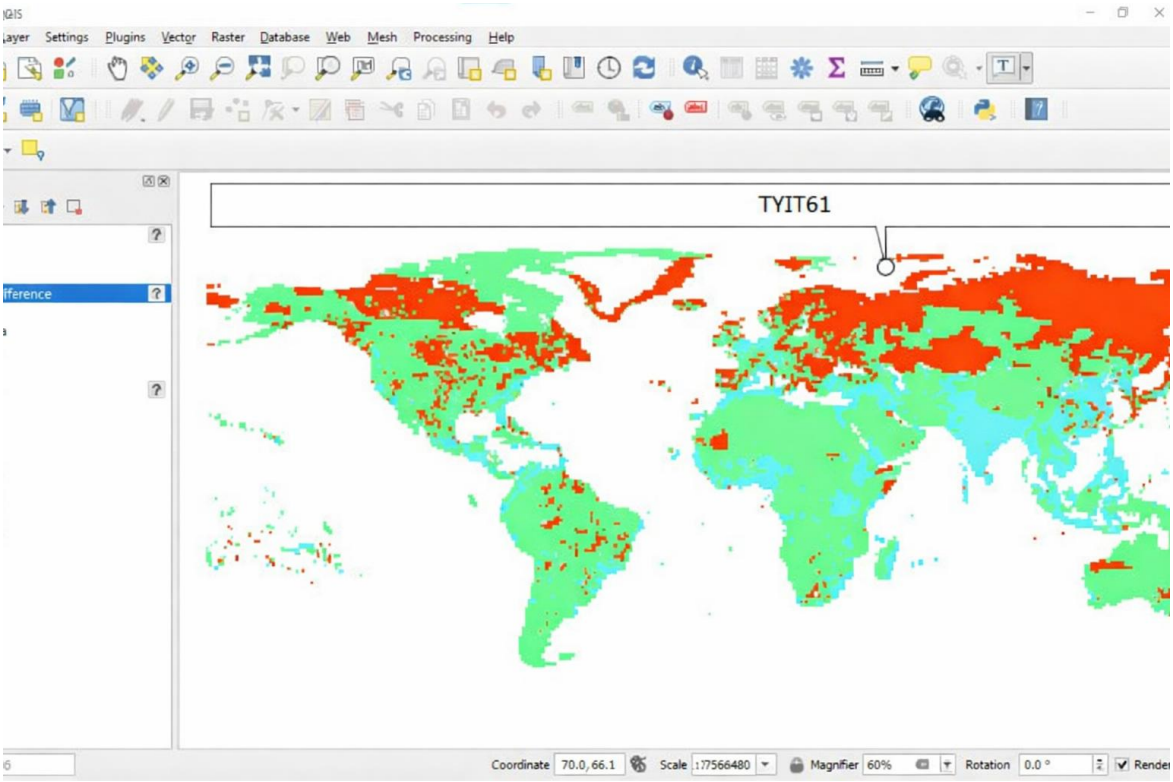
- Select Spectral from the Color ramp menu.
- Remove all existing classifications.
- Add required classes using the + and – buttons.
- Click Apply, then click OK





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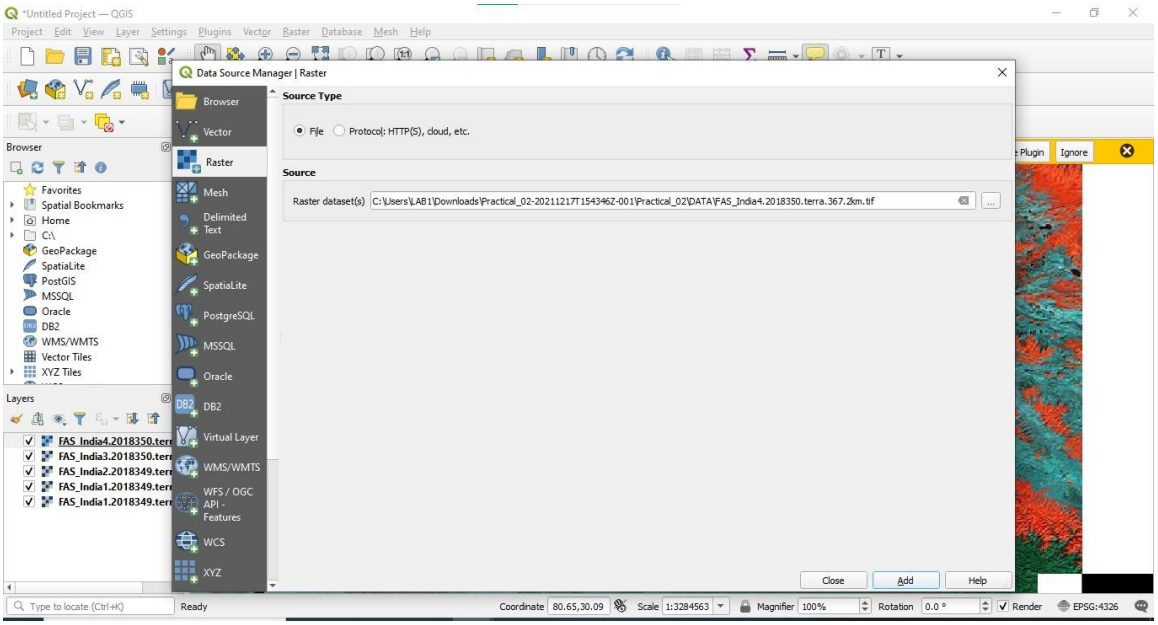
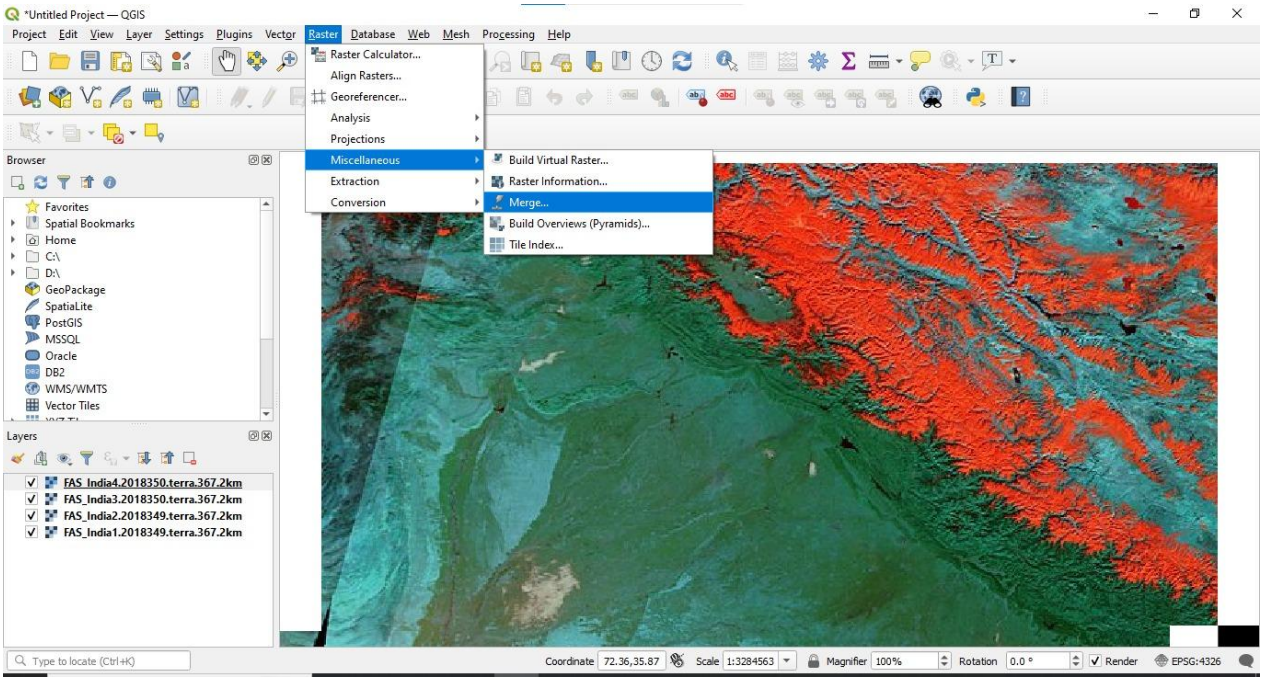
C. Mosaicking & Clipping

1. Go to Layer -> Add Layer -> Add Raster Layer -> Add 4 .tif images.



2. Go Raster -> Miscellaneous -> Merge



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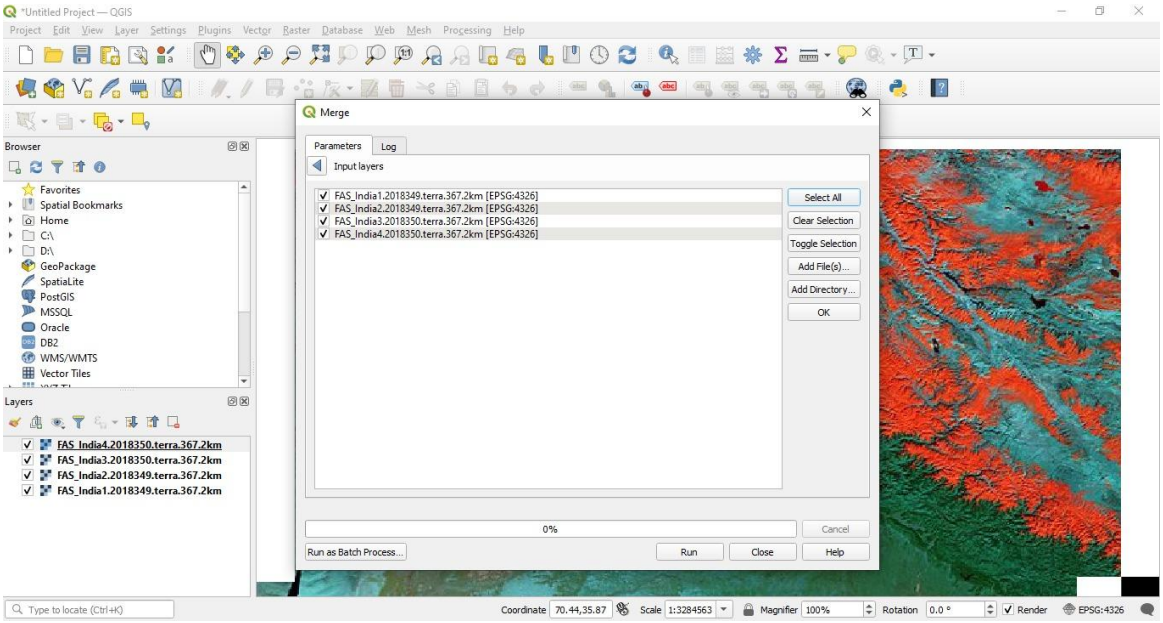
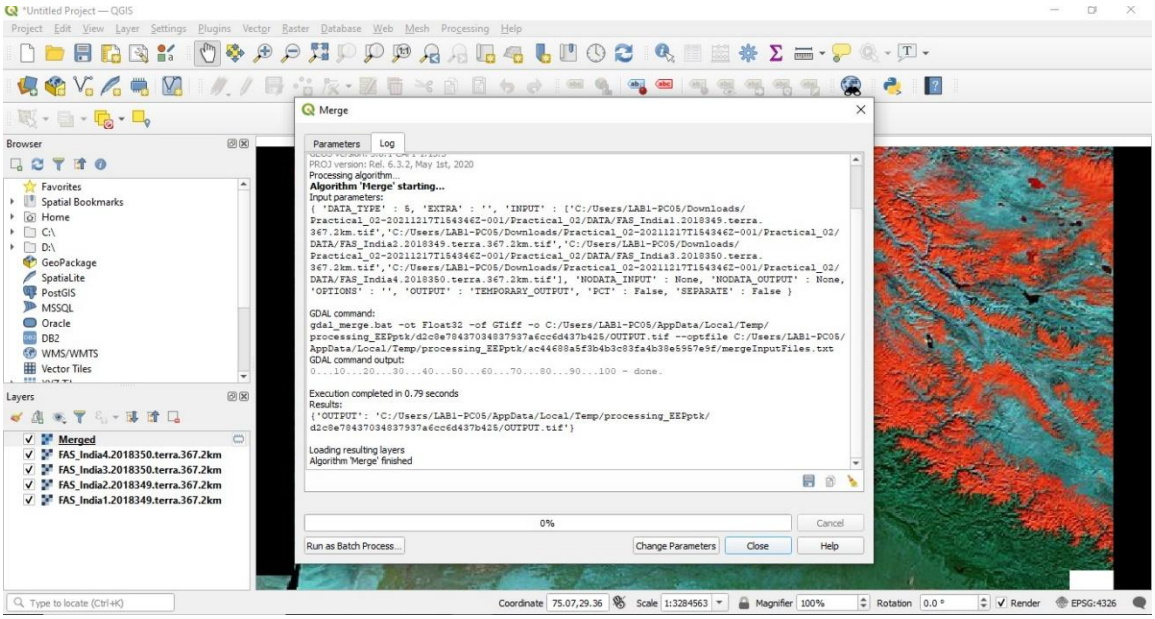
3. In the Merge dialog box:
   - Select all the input raster layers to be mosaicked.
   - Specify the output file name and choose the location to save the merged image.
   - Click the Run button to start the mosaicking process.
   - After successful execution, click Close.





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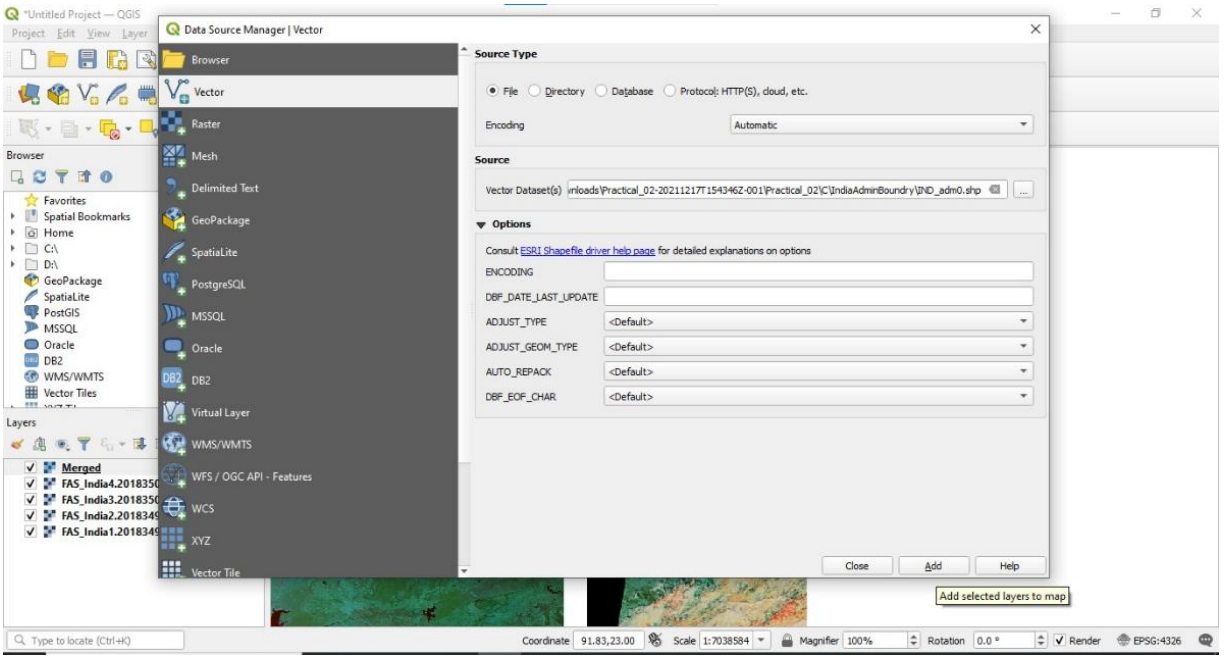
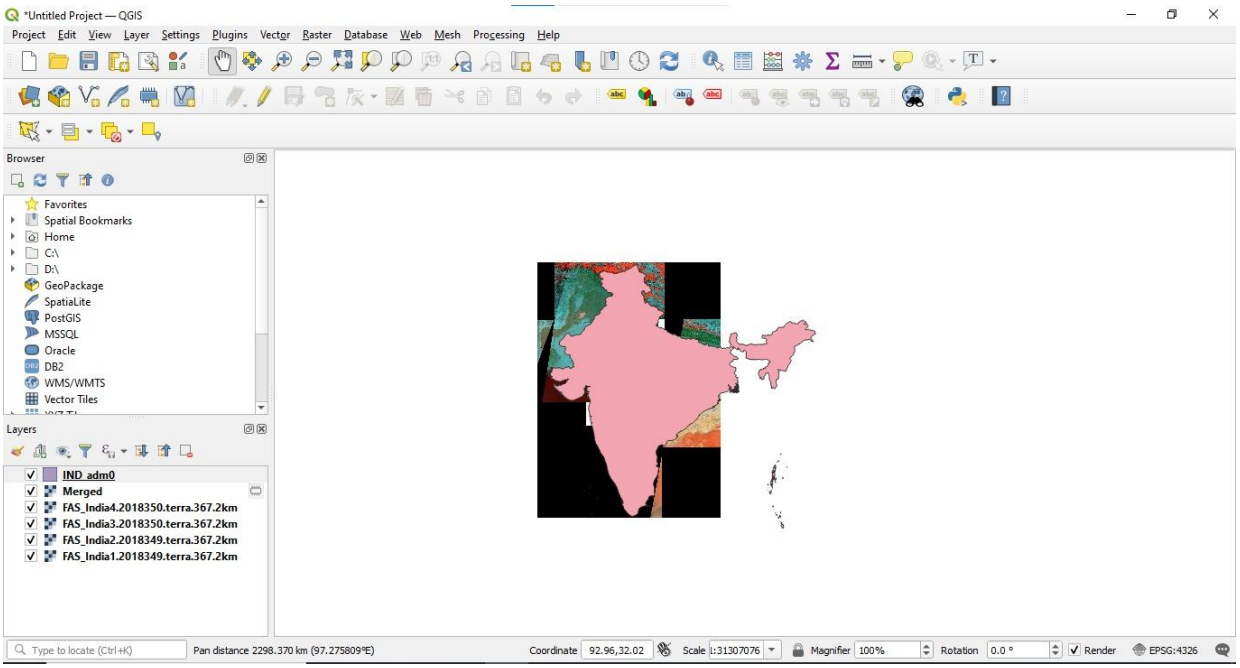
4. Go **to** Layer → Add Layer → Add Vector Layer -> Browse and select IND_adm0.shp from the folder Practical_No_2_Data\IndiaAdminBoundry -> Click Add, then click Close.





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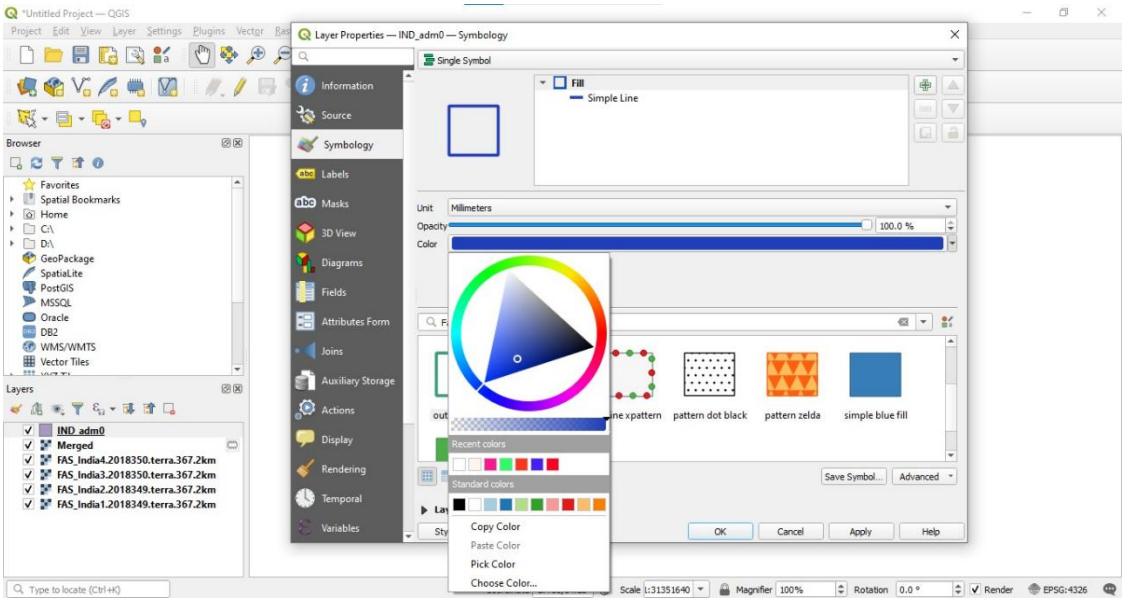
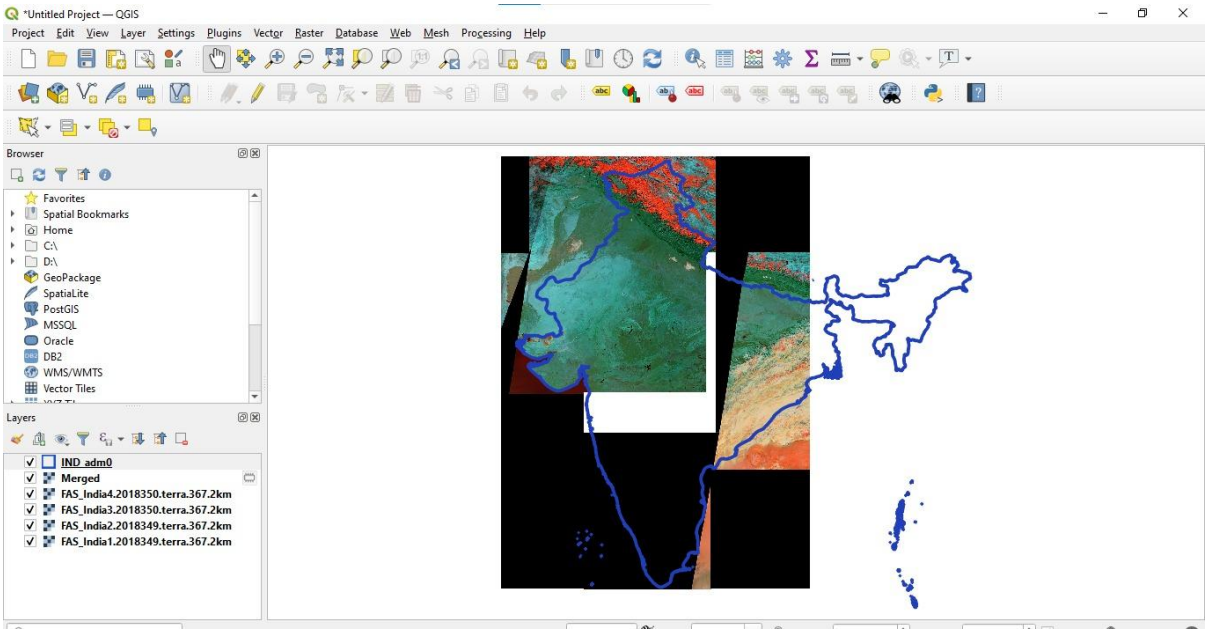
5. From Layer Properties -> Select Symbology -> Select any one of the following then click Apply and ok Buttons.





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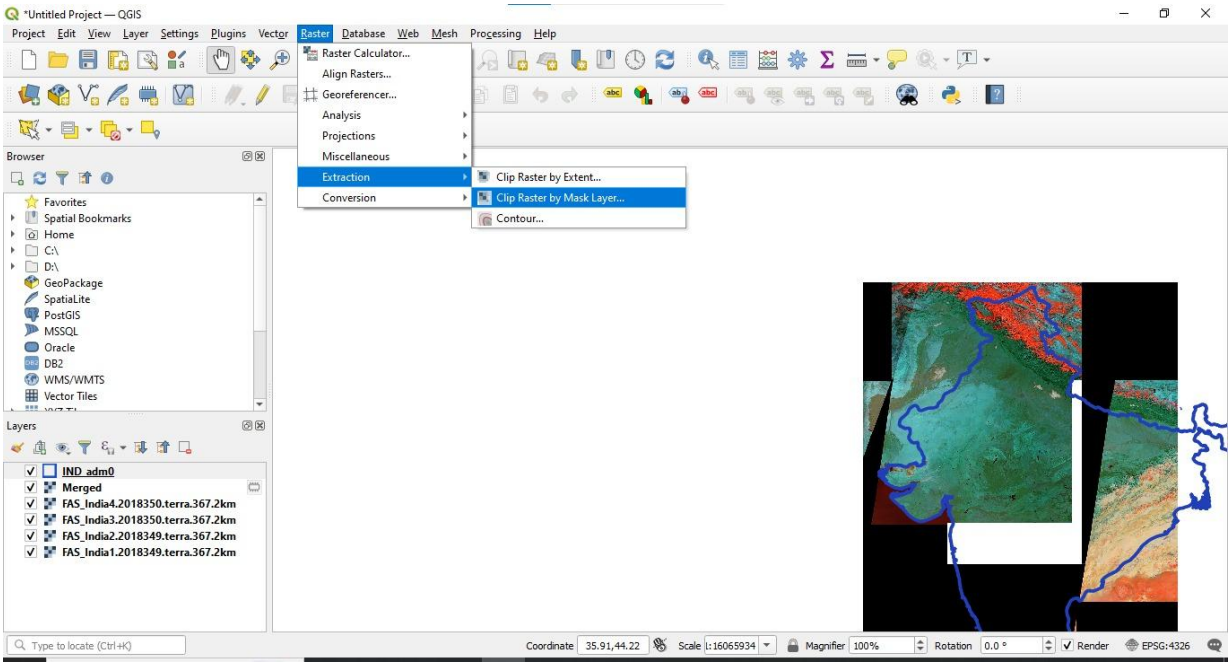
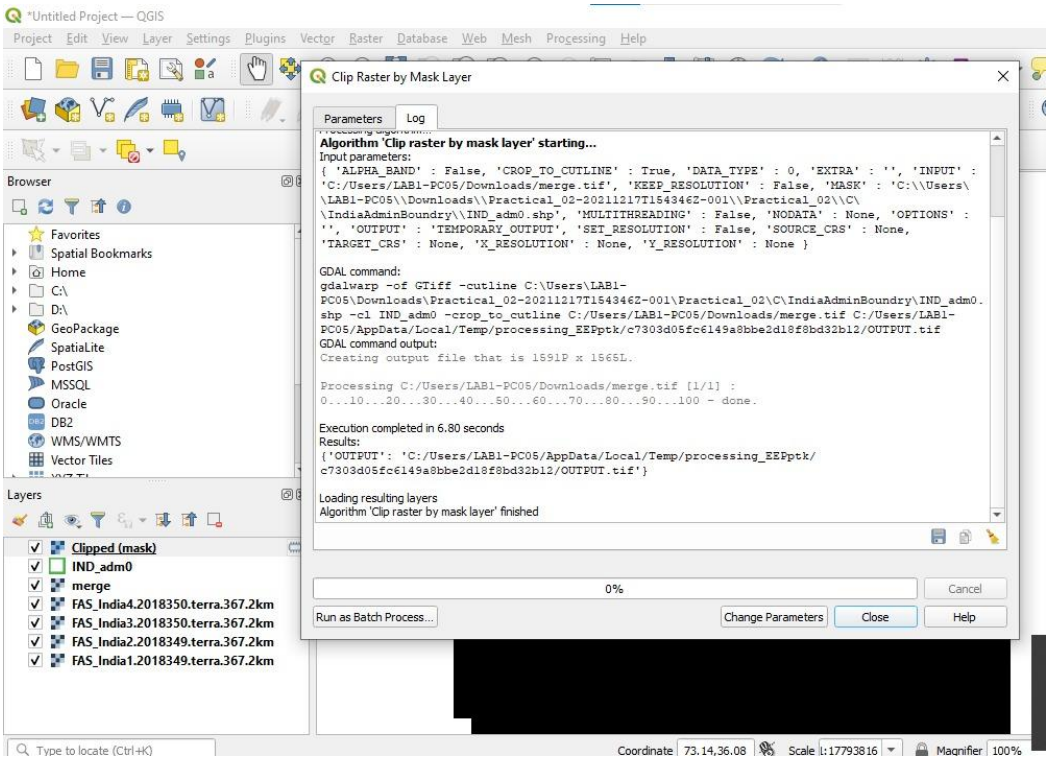
6. Go to Raster -> Extraction ->Clip Raster by Mask Layer



7. Select the merge raster layer as input and Ind_adm0 as mask layer



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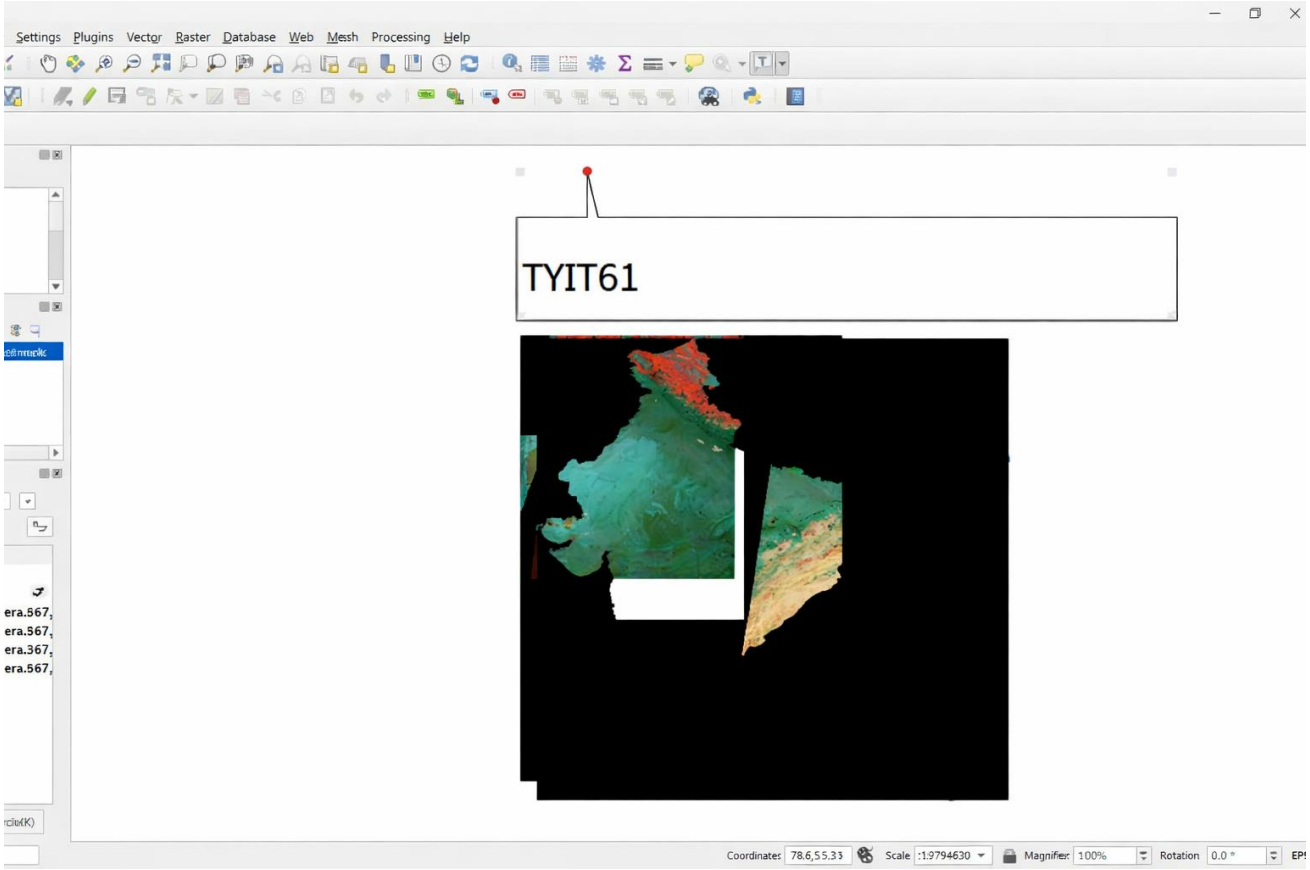

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